

Yonghao Shi

Ph.D. Student in Computer Science, Simon Fraser University

Human-Computer Interaction | Smart Environments | Computational Materials | Applied AI

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PROFILE

My research lies at the intersection of Human-Computer Interaction, smart environments, computational materials, sensing systems, and applied AI. I design hardware-enabled interactive systems that turn everyday materials, especially wood panels, furniture, and room-scale structures, into sensing and actuation interfaces. My current work focuses on sustainable computational wood, battery-free wireless sensing, RFID-augmented environments, low-barrier fabrication, and user-centered evaluation. Before joining SFU, I worked as a software engineer and project manager, which shaped my ability to build reliable systems and manage research prototypes end to end.

PUBLICATION

- [C4] **3Duino: A Low-Barrier Platform for Prototyping Interactive 3D-Printed Devices.**
Yonghao Shi, Zejia Cai, Liuyi Chen, Yuning Su, Liang He, Xing-Dong Yang, and Te-Yen Wu.
The 10th Annual ACM Symposium on Computational Fabrication (SCF 2025). DOI: [10.1145/3745778.3766649](https://doi.org/10.1145/3745778.3766649)
- [C3] **Woodowl: Electrode Isolation for Electromagnetic Shielding in Triboelectric Plywood Sensors.**
Yonghao Shi, Chenzheng Li, Yuning Su, Xing-Dong Yang, and Te-Yen Wu.
Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI 2024). DOI: [10.1145/3613904.3642304](https://doi.org/10.1145/3613904.3642304)
- [C2] **Tagnoo: Enabling Smart Room-Scale Environments with RFID-Augmented Plywood.**
Yuning Su, Tingyu Zhang, Jiuen Feng, Yonghao Shi, Xing-Dong Yang, and Te-Yen Wu.
Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI 2024). DOI: [10.1145/3613904.3642356](https://doi.org/10.1145/3613904.3642356)
- [C1] **Laser-Powered Vibrotactile Rendering.**
Yuning Su, Yonghao Shi, Da-Yuan Huang, and Xing-Dong Yang.
Adjunct Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology (UIST Adjunct 2023). DOI: [10.1145/3586182.3615795](https://doi.org/10.1145/3586182.3615795)

WORK EXPERIENCE

- 2020 - 2021 **Project Manager**, Institute of Information Engineering, Chinese Academy of Sciences.
Managed software and data-oriented engineering projects, coordinated timelines, and supported cross-functional delivery.
- 2018 - 2020 **Software Engineer**, Institute of Information Engineering, Chinese Academy of Sciences.
Worked on backend systems, data infrastructure, and software development for research and engineering applications.
- Jun. - Sep. 2017 **Software Engineer**, PwC Beijing.
Contributed to software engineering tasks in a professional consulting and technology environment.

EDUCATION

- 2022 - Present **Ph.D. in Computer Science**, Simon Fraser University, Burnaby, Canada.
Research: Human-Computer Interaction, smart environments, and computational materials.
- 2017 - 2018 **M.Sc. in Computer Science**, University of Plymouth, United Kingdom.
Thesis: Data Science and Business Analysis.
- 2016 - 2017 **B.Sc. in Electrical and Electronic Engineering**, University of Plymouth, United Kingdom.
- 2013 - 2016 **B.Sc. in Electrical and Electronic Engineering**, Beihang University, China.